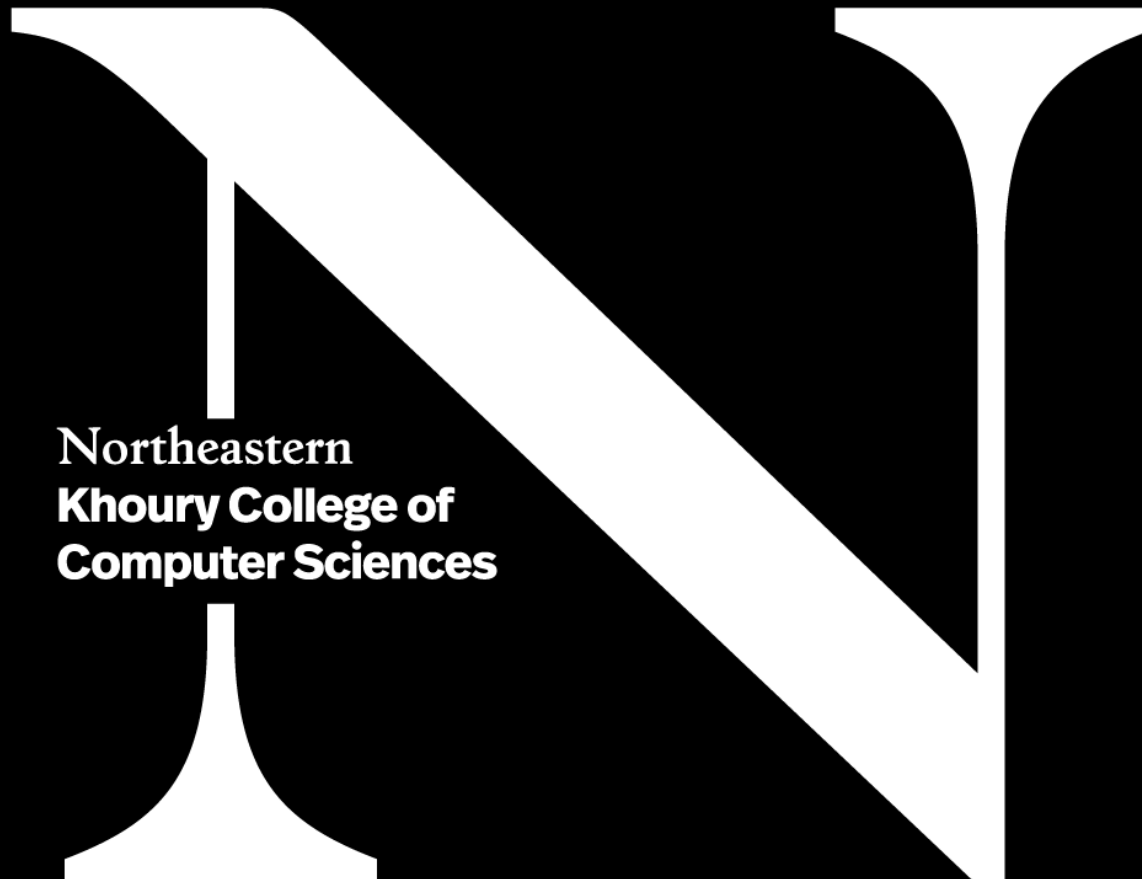
PyTorch Primer

CS 6120: Natural Language Processing



What is PyTorch?

- Open-source machine learning library for Python
- Used for applications such as computer vision and natural language processing
- Simplifies using CUDA for deep learning models.

Getting started

- Python Version: 3.10.x
- Pytorch Version: 2.2.x
- IDE
  - VSCode
  - Anaconda Jupyter Notebook

Installing PyTorch

- Using pip

  pip install torch torchvision torchaudio

- Using conda

  conda install pytorch torchvision torchaudio -c pytorch



Verify Installation

```
PS C:\Users\anjit> py
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023, 00:38:17) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> print(torch.__version__)
2.2.2+cu118
>>>
```

What is the use?

- Helps with the flow of making the model
  - Data preparation
  - Model Definition
  - Loss Function and Optimizer
  - Forward and Backward pass

Key Libraries

- torch: Core library for tensors, tensor operations, and autograd.
- torch.nn: Provides neural network layers and utility functions to build models.
- torch.optim: Optimization algorithms for training models (e.g., SGD, Adam).
- torchvision: Common image datasets (like MNIST, CIFAR-10) and pre-trained models.



What is a tensor?

- A tensor is a multi-dimensional array used in PyTorch
- It can store
  - Single value
  - 1 Dimensional value
  - 2 Dimensional value
  - n-Dimensional value
- It is designed to be flexible and easily integratable with GPUs.



Implementation

```python
import torch
import torch.nn as nn
import torch.optim as optim

# Implement a simple neural network
class SimpleNN(nn.Module):
    def __init__(self):
        super(SimpleNN, self).__init__()
        self.fc1 = nn.Linear(784, 128)
        self.fc2 = nn.Linear(128, 10)

    def forward(self, x):
        x = torch.relu(self.fc1(x))
        x = self.fc2(x)
        return x

model = SimpleNN()
criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(model.parameters(), lr=0.01)

print(model)
```

[1] ✓ 2.5s

```
SimpleNN(
  (fc1): Linear(in_features=784, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=10, bias=True)
)
```

Jupyter Notebook

https://bit.ly/cs6120-pytorch-primer

NORTHEASTERN